

Software Requirements Specification for a Collaboration Platform for Project Studies

Report for Project Study

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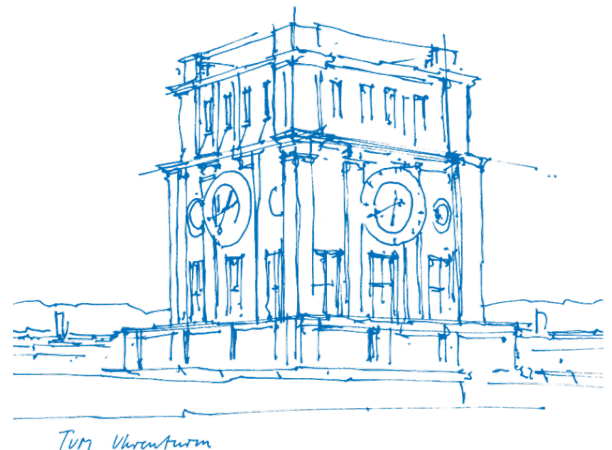
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1. Introduction

1.1. Purpose of this Document

This specification defines the requirements for a project studies platform, developed as part of a bachelor's Project Study at TUM Campus Heilbronn in collaboration with BirdVision. It is adapted from ISO/IEC/IEEE 29148, scaled down and extended with a literature review section to meet the academic requirements of a project study report. Every requirement in Sections 6 and 7 is traceable back to its source and forward to its prototype coverage, per Section 9. That traceability, more than any single requirement, is what this document is meant to demonstrate.

1.2. Project Background and Motivation

A project study for TUM School of Management students is a research or practical project, carried out by a student team in collaboration with a company and a supervising professorship. This project, "Project Studies Platform for BirdVision", was assigned to build a platform supporting a project study process similar to this one, which this document itself is a product of.

Companies, professors, and students currently coordinate project studies at TUM Campus Heilbronn in three ways: university staff sourcing companies directly and sending out given topics to professors, professors and companies agreeing between themselves, or students agreeing with a company and finding a supervising professor on their own. These processes are decentralized and realized via a variety of mediums. Recurring pain points arising from the status quo were identified through semi-structured interviews, which this document translates into a scoped set of requirements.

1.3. Scope of the Platform

The scope of the platform is limited to the project study process at TUM Campus Heilbronn, rather than industry-academia collaborations in general. The reasoning behind this scope and discussion on how this platform can be adapted for different institutions is written out in sections 4.4 and 4.5. Broader findings from the interviews are not excluded and presented in section 5.

To deliver on the project goal of a testable MVP, this document is accompanied by a non-functional prototype.

1.4. Definitions, Acronyms, and Abbreviations

SRS Software Requirements Specification

FR / NFR Functional Requirement / Non-Functional Requirement

MoSCoW Prioritization scheme: Must, Should, Could, Won't (Section 6.1)

NDA Non-Disclosure Agreement

GDPR General Data Protection Regulation

1.5. Document Overview

Section 2 reviews the literature behind the project's methodological choices. Section 3 goes over the interview methodology, sample, and its limitations. Section 4 describes the platform's overall scope and boundaries. Section 5 presents the pain points the requirements are derived from. Sections 6 and 7 specify the functional and non-functional requirements. Section 8 provides system models. Section 9 traces requirements back to their source and forward to prototype coverage. Section 10 discusses risks and conclusions.

2. Literature review

In this section we review the academic grounding for the project's methodological choices. In particular, the requirements engineering approach and the broader phenomenon of academia-industry collaboration that the platform aims to address.

2.1. Requirements Engineering Grounding

2.1.1. Selection and Adaptation of the Requirements Engineering Standard

This specification follows ISO/IEC/IEEE 29148:2018 (Society, 2018) – the current international standard for requirements engineering, which supersedes IEEE 830-1998 (Society, 1998). The predecessor addressed the specification document alone, whereas 29148 covers the requirements process as a whole, including elicitation, analysis and validation. Consequently, such a form is more appropriate for this project, since much of what follows concerns how requirements were derived from stakeholder interviews and how strongly each is supported. The standard is applied in scaled-down form appropriate to a single-semester project and extended with a literature review section to meet the academic requirements of a project study report.

2.1.2. Interview-Based Requirements Elicitation

Semi-structured interviews were used to explore stakeholder needs without assuming predefined system requirements. Ferrari et al. (2022) show that interviews support an iterative elicitation process in which stakeholder initial ideas are progressively clarified and refined into requirements through the dialogue and iterative feedback. Accordingly, this project first analyzed interview data inductively to identify recurring needs and pain points, which then served as the basis for deriving platform requirements.

2.1.3. Sample Adequacy and Thematic Saturation

MacQueen et al. (1998) defined thematic saturation as the point at which no new codes emerge from additional interviews. And, the number of interviews required to reach the point of saturation is a recurring question in qualitative research.

Guest et al. (2006) found that little new information emerged after the first 12 of their 60 interviews.

Hennink et al. (2017) argue that the answer depends on what saturation is taken to mean. No new codes appeared in their data after 9 interviews, but arriving at a rich

understanding of those codes took between 16 and 24. These figures come from fairly uniform groups of participants.

Hagaman & Wutich (2017) worked across 4 sites and found that 16 interviews or fewer sufficed within a single group, while themes running across groups needed 20 to 40.

These thresholds provide a basis for assessing the 26 interviews conducted for this study rather than merely reporting their number. The student group, at 13 interviews, exceeds the code-saturation threshold reported by Hennink et al. and falls within the range Hagaman and Wutich associate with thematic convergence in a single group; the pain points reported for students in Section 5.1 can therefore be regarded as reasonably stable. The company, professor and university-staff groups, at 5, 6 and 2 interviews respectively, fall below every threshold cited above, and findings specific to them are correspondingly treated as indicative. The cross-cutting findings in Section 5.4, which draw on the full sample of 26, sit within the range associated with cross-group themes, though not at its upper end. Requirements are accordingly framed throughout as informed proposals rather than validated findings.

2.1.4. Requirements Traceability

Requirements traceability is the ability to follow a requirement forward into design and implementation and backward to the source it came from. Gotel & Finkelstein (1994), drawing on interviews and questionnaires with over 100 practitioners, distinguish these two directions and argue that the backward one is where projects usually fail.

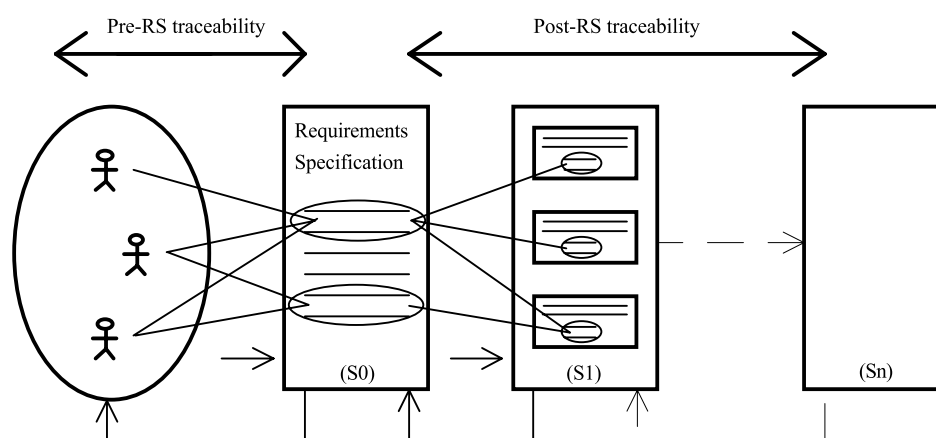


Figure 1: Illustration of the two directions of requirements traceability from Gotel & Finkelstein (1994)

Post-specification traceability, linking requirements to what was built from them, is comparatively well supported by tools. Pre-specification traceability, linking a requirement back to the stakeholder statement and rationale that produced it, tends to be neglected, and most of the problems attributed to poor traceability turn out to originate there.

Section 9 is organised along both directions: Section 9.1 provides the pre-specification record, tracing each requirement back to its pain point and interview material, while Section 9.2 and 9.3 provide the post-specification record, mapping requirements forward to prototype screens and accounting for those left unimplemented.

2.1.5. Requirements Prioritization

Prioritization techniques differ mainly in how much they demand of the team using them. Reviewing the field, Achimugu et al. (2014) find that pairwise methods such as the Analytic Hierarchy Process become impractical as requirement counts grow, and that cost-value approaches depend on effort and benefit estimates unavailable at this stage of a project. MoSCoW asks for neither, sorting requirements into four ordinal classes on judgement alone, which suits a small team working to a fixed deadline on requirements that have not been validated. The criteria used for that sorting are set out in Section 6.1.

2.2. Academia-Industry Collaboration

2.2.1. Collaboration friction as an established phenomenon

Collaboration between universities and companies has been studied extensively, and the difficulties it produces are documented rather than incidental. Ankrah & AL-Tabbaa (2015) synthesise a fragmented body of work into a process framework covering why the two sides collaborate, the forms collaboration takes, and the organisational obstacles that recur across national contexts. Coordination and communication problems appear throughout that literature, independent of country, discipline and collaboration format. Since this problem is documented across many institutions, the difficulties reported in Section 5 are treated as a specific case of it rather than as a local peculiarity.

2.2.2. Barriers and information asymmetry between stakeholder groups

Bruneel et al. (2010) separate two kinds of barrier to university-industry collaboration. Orientation-related barriers arise from differences in what each side is trying

to achieve and on what timescale. Transaction-related barriers arise from the mechanics of working together, including administration and intellectual property. Prior collaboration experience reduces the first kind, while trust between the parties reduces both. Difficulties such as identifying the right contact person or establishing what expertise is available fall into the second category, which is where a platform can plausibly intervene. This distinction is used in Section 5.4 to organise the pain points that recur across stakeholder groups.

2.2.3. Divergent motivations of academic and industry partners

The two sides enter collaboration for different reasons. Perkmann et al. (2013) show that academic engagement is driven largely by research-related motives, including access to data, funding and problems worth studying, while firms pursue capability, talent and commercial application. Neither set of motives is illegitimate, but they are not automatically aligned, and the semester rhythm that governs university work rarely matches company project timelines. A platform mediating between the two therefore cannot assume a shared objective; it has to make each side's constraints legible to the other. Section 5.4 returns to this point where the mismatch surfaced in the interview data.

2.2.4. Platforms and intermediaries as a category of solution

Where two parties struggle to find and evaluate each other, a third party can lower the cost of doing so. Howells (2006) develops a typology of innovation intermediaries and defines them as bodies acting as brokers between parties in the innovation process, performing functions such as scanning, matchmaking and coordination. The platform specified in this document sits in that category, with the difference that it serves three groups rather than two.

3. Research Approach and Limitations

3.1. Interview Methodology

The study followed a sequential design. The requirements engineering literature reviewed in Section 2.1 informed the interview guide, and a scoping discussion with BirdVision fixed what the client needed from the study. Stakeholder groups were then defined, participants recruited and interviews conducted. Responses were consolidated into a single matrix, analysed by group, and translated into requirements, from which the prototype, the traceability table, the use case diagrams and the key user flows were produced.

Interviews were semi-structured, using a common guide across all groups so that responses stayed comparable, with follow-up questions free to depart from it wherever a participant raised something unanticipated. The guide covered four areas:

1. **Current collaboration experience** — what forms of collaboration the participant had taken part in, and how they found them.
2. **Motivation** — what makes collaboration worth entering, and what makes a project attractive.
3. **Problems and barriers** — what goes wrong during collaboration, and what prevents people from engaging at all.
4. **Platform and solution needs** — asked last, so that earlier answers were not framed around a software solution.

Only the last area concerns the platform directly. The first three establish how collaboration currently works and why it is difficult, and are reported in Section 5 independently of the specification.

Four stakeholder groups were distinguished: students, companies, professors and university staff. Staff were separated from professors because their role is administrative rather than academic, and their pain points proved to differ accordingly. Participants were recruited through campus contacts and, for most company representatives, at Career Factory; Section 3.2 discusses the resulting composition and limitations.

Responses were recorded in a matrix with participants as columns and the four areas as rows, allowing each theme to be read across all participants and across groups. Section 3.3 describes how the requirements were derived from it.

3.2. Sample Composition and Limitations

Twenty-six semi-structured interviews were conducted across four stakeholder groups. Appendix B lists all participants; the distribution is summarised below.

Group	n	Institutional background
Students	13	9 TUM Heilbronn, 2 TUM Garching, 1 RWTH Aachen, 1 Reutlingen
Companies	5	Startup, industrial manufacturing, software, energy management, IT services
Professors and researchers	6	2 TUM Heilbronn, 1 TUM Munich, 1 RWTH Aachen, 1 Koblenz, 1 Passau
University staff	2	2 TUM Heilbronn

Table 1: Interview sample by stakeholder group

The sample spans five universities. Eight of the 21 university-affiliated participants are based outside TUM Campus Heilbronn, so where the same difficulties recur across institutions, they are unlikely to be local artefacts.

Two limitations shape how the company findings should be read. Most company representatives were recruited at Career Factory, a university–company event on the Bildungscampus, so the sample includes no firms that have never collaborated with a university or have stopped doing so. More importantly, four of the five hold roles in human resources, apprenticeship management or campus relations. Their interest in the university is oriented towards recruiting, and the requirements derived from this group reflect that. Needs around research cooperation, confidentiality and intellectual property are under-represented. The founder interviewed as C1 offers a partial counterweight, having neither a recruiting function nor an employer-branding budget, and is treated in Section 6 as a contrasting case rather than as one observation among five.

The company, professor and staff groups are small, and findings specific to them are treated as indicative. The staff group is smallest at two, though both are the administrators who currently run the project-study process at TUM Campus Heilbronn, so the group covers the relevant population rather than a sample of it.

This background of existing engagement is unevenly distributed across the sample. Several students and professors interviewed had no established route into univer-

sity–industry collaboration, so their accounts include the perspective of those outside it. All five company representatives, by contrast, were already engaged with universities at the time of the interview. The company data therefore describes how engagement works for firms that have already committed to it, and says little about what would bring in a firm that has not.

3.3. From Interview Insights to Requirements (how findings were translated, requirements framed as informed proposals, not validated findings)

The analysis started from the response matrix described above. Each theme was read across all 26 participants, first within a stakeholder group and then across groups, and recurring statements were grouped into pain points. Pain points named by only one group are reported under that group in Section 5, while those raised by more than one group are reported once among the cross-cutting findings at the end of that section. Each pain point lists the participants who raised it, so a reader can see how many independent sources stand behind it.

Requirements were then derived from these pain points. The two sets do not match one to one. Some pain points lie outside what a platform can fix, and those are accounted for in Section 9. Some requirements, in turn, come from elsewhere. Each requirement therefore states what kind of evidence it rests on:

Basis	Meaning
Interview	Taken from one or more interview statements, cited by participant ID.
Process documentation	Taken from how the project-study process currently works at TUM Campus Heilbronn, rather than from interview data.
Design decision	A choice made by the authors, where neither source supplied a requirement. Recorded as Team Decision in the traceability table (Appendix C), with the reasoning given at the requirement itself.

Table 2: Evidence basis recorded for each requirement

Three requirements rest on a design decision rather than on interview data or the current process: Student Profiles (FR-9), Dual Approval (FR-15), and Language Support (NFR-3). Each is argued for where it is specified, and FR-15, which proposes a change to how student selection currently works, is discussed again in Section 10.

Two further attributes are recorded and feed the prioritization scheme in Section 6: whether the underlying pain point recurs across groups, and whether the requirement depends on TUM's administrative arrangements or would apply at any university.

The prototype was built from the requirement set as a whole rather than requirement by requirement. The mapping from requirement to screen in Section 9 was therefore drawn up after implementation, not before it.

Two limits follow. The requirements were never taken back to participants for confirmation, and the step from interview statement to pain point rests on the authors' reading of the material. Together with the sample limitations described above, this is why the requirements in Sections 6 and 7 are presented as informed proposals. Each one can be traced to its evidence, and the strength of that evidence is recorded, but none has been checked against the people it came from.

4. Overall Description

4.1. Product Perspective

The platform is a new, self-contained system rather than an extension of existing university software. It acts as an intermediary among parties who currently find one another through scattered institute pages, personal contacts, job boards and events, and its purpose is to reduce the cost of that search and standardise the information exchanged once contact is made.

What exists at the end of this project study is a prototype. It demonstrates the interfaces and flows specified here, but implements no persistent storage, no authentication against university identity systems and no integration with existing institutional processes.

4.2. Stakeholder Groups, Roles and Motivations

- **Students:** search for projects, apply and carry out the work, motivated by practical experience, industry contact and, in several cases, employment prospects. Their engagement is conditional — several said they would withdraw from a project seen as serving only a commercial interest.
- **Companies:** post projects and receive results, motivated by talent contact, access to expertise and usable outcomes. The companies interviewed were oriented mainly towards recruiting, and the requirements derived from them reflect that.
- **Professors and academic researchers:** supervise and assess projects, motivated by access to real problems and by teaching value. Their constraint is time.
- **University staff:** administer the process, matching requests to supervisors and tracking progress. Their main obstacle is obtaining timely responses from professorships.

These motivations are not aligned by default. The platform does not assume they are; its function is to make each group's constraints visible to the others.

4.3. Assumptions and Constraints

The specification assumes a university runs the platform, not a commercial third party. This matters because several requirements depend on the operator being able to confirm that a profile really does belong to a member of the university, and only the university itself holds that information. We further assume that every user has

an affiliation which can be verified, and that each project study runs to a timeline agreed between the supervising professor and the company rather than to a fixed institutional calendar.

There is one more assumption, about how selection currently works. Students at the TUM Global Center for Family Enterprise are chosen by the university alone, with no involvement from the companies whose projects they apply to. Section 6 contains a requirement that would change this. We have flagged it there as a proposed change to the process rather than presenting it as something the platform merely automates.

Our constraints were those of a single-semester project study, with no budget and a small team. Everything here rests on interview data. We could not consult institutional process documentation beyond what participants described to us, and nobody with legal expertise reviewed the data-protection and confidentiality questions that operating such a platform would raise.

4.4. Scope Boundary: TUM-Specific Focus (explicit justification: administrative/collaboration processes differ significantly across universities, so single-institution depth was prioritized over shallow generality)

The platform is specified against one process in particular: the project study process at TUM Campus Heilbronn. Administrative arrangements vary a great deal between universities, and a specification broad enough to fit all of them would be too vague to build from. We chose depth against a process we could examine closely.

The problems it addresses are another matter. We modeled a local workflow, but the difficulties it responds to came up at every university in our sample.

4.5. Transferability to Other Institutions (brief discussion of what would need to adapt for other universities)

Each requirement is tagged according to how far it depends on local arrangements:

Tag	Meaning
Generalizable	Would apply at any university.
Adaptable	Depends on an arrangement other institutions also have in some form; needs reconfiguring rather than redesigning.
Institution-specific	Tied to how TUM Campus Heilbronn organises the process; would require rework elsewhere.

Table 3: Transferability tags applied to requirements

These tags feed the prioritization scheme in Section 6.1: requirements that are generalizable and grounded in cross-cutting pain points rank above those that are institution-specific or supported by a single group.

4.6. Out-of-Scope Items (e.g. backend functionality not prototyped)

Outside the scope of this specification and the prototype: persistent storage and backend services, authentication against university identity providers, automated student–project matching, contract and intellectual-property handling, integration with university administration systems, and how the platform would acquire its first users on each of the three sides.

5. Stakeholder Needs (Pain Points)

The pain points in this chapter are derived from semi-structured interviews conducted across three primary stakeholder groups relevant to the project study process. Each pain point is presented with the interview ID it was sourced from. See appendix 11.B for anonymized interview counts per stakeholder group and matching interview IDs.

BirdVision's primary interest with this project study was gaining insight into industry-academia collaborations as a whole, particularly the stakeholders' motivations and the problems they encounter. The interviews were accordingly conducted on the broader premise of industry-academia collaborations, in order to gather as wide a range of insights as possible. However, as the stated aim of the project is a platform supporting project studies specifically, and the scope of the platform is limited, not every pain point identified applies directly to the project study process, or to TUM in particular. These findings are nonetheless documented in full below.

5.1. Student Pain Points

5.1.1. Discoverability

Students consistently report that there is no clear, centralized listing for company projects. Project opportunities are described as scattered across fragmented institute websites or buried in static job boards (S6, S7, S8).

5.1.2. Missing or insufficient listing information

Project information about workload, technical requirements, deliverables, and whether the company genuinely prioritizes the project's outcome is frequently unavailable before applying (S8, S9). Company name, expected outcomes, and required skills were independently named as the minimum information needed to make an informed decision (S7, S9).

5.1.3. Communication and responsiveness

Slow communication loops and vague project scope were named as reasons a student would abandon an application outright (S8). Separately, students report response delays during the application process and a lack of feedback following rejection (S12, S13).

5.1.4. Language barrier

German-language requirements were named as a disadvantage for international students on multiple, independent occasions (S5, S7, S10).

5.1.5. Team formation

A student named the inability to find a team as something that would stop them from applying altogether, and separately requested an automated team-matching feature (S10).

5.1.6. Application process friction

One student described having to repeatedly re-enter the same information across each company's own application platform, alongside generic, under-informed initial screening and a lack of feedback after rejection (S1). A separate student reported that project details were withheld pre-NDA, leading to a fundamentally different understanding of the task once disclosed (S9). Yet another student described the challenges they faced in finding an academic supervisor for their company project (S12).

5.1.7. Value and goal alignment

Several students indicated they would disengage from a project perceived as serving only the company's commercial interest without genuine educational or research value (S6, S8). This pattern recurs across other stakeholder groups and is developed further in Section 5.4.

5.2. Professor/University Pain Points

5.2.1. Information standardization

Professors report that companies frequently fail to provide sufficiently detailed project briefs, desired area of expertise, objective, and data availability, leaving task scope unclear for both supervisor and student (P3, P4, P6).

5.2.2. Contact and coordination burden

Locating appropriate contacts and coordinating a collaboration was named as time-consuming and requiring substantial advance planning (P1). On the administrative side, obtaining a timely response from professorships regarding supervision requests was named as the primary obstacle encountered by university staff coordinating intake (U1, U2).

5.2.3. Students' engagement with supervisors during the project

One professor observed that students frequently under-communicate with their academic supervisor once a company relationship is underway, directly linking this

to reduced outcome quality, and has introduced mandatory monthly check-ins as an informal mitigation (P4).

5.2.4. Outcome measurement

Two different professors reported that there is no formal mechanism for assessing whether the effort invested in a project study was proportionate to its value. One describes this as entirely unmeasured, the other as narrative rather than KPI-based (P1, P5). This pattern recurs in Section 5.4.

5.2.5. Resource pressure

Time and financial resources dedicated to these collaborations were named as the first budget item companies tend to cut, constraining the professor's ability to sustain partnerships (P5).

5.3. Company Pain Points

5.3.1. Contact identification and structural complexity

Identifying the correct contact person and aligning expectations, timelines, and available resources was the most frequently cited difficulty (C3, C4). University structures were separately described as complex and inconsistent between institutions (C4), directly reinforcing the transferability discussion in Section 4.4-4.5.

5.3.2. Missing information when searching for partners

Companies report a lack of visibility into contact details, areas of expertise, ongoing projects, and collaboration opportunities (C4), as well as the absence of a clear overview of available posting/collaboration options (C5).

5.3.3. Output quality and delivery friction

For one company, the central concern was not access to universities but the practical usability of delivered outcomes, compounded by software-tooling mismatches between student and company environments (C2). A second company reported an inability to assess student or study quality in advance, which discourages deeper engagement. (C5).

5.3.4. Need for trust and verification

Companies want assurance that student and professor profiles are genuinely affiliated with their stated institution, citing existing third-party verification services elsewhere as a reasonable precedent (C4).

5.3.5. Screening signal degradation

One company noted that student CVs are increasingly perceived as AI-generated and therefore less reliable as a screening tool, expressing a preference for more direct channels, including contact with student clubs, over formal university structures (C3).

5.4. Cross-Cutting Findings (pain points recurring across all three groups)

5.4.1. Contact-person and reachability breakdown

Identifying and reaching the correct person was named independently by professors (P1), companies (C3, C4), and university staff (U2, U3). It is the most broadly and heavily evidenced finding in the dataset, spanning three stakeholder groups.

5.4.2. Bidirectional information insufficiency

Every stakeholder group reports the same underlying gap directed at a different counterpart: students want more complete project information from companies and professors (S7, S9); professors want companies to provide more structured project detail (P3); companies want more visibility into student and professor profiles, expertise, and ongoing activity (C4). The specific direction differs but the pattern recurs across all three groups.

5.4.3. Academia-industry goal misalignment

This is the most substantively supported cross-cutting finding. The university research staff perspective names it directly (U1), one company frames it as their central complaint about output usability (C2), a second names it explicitly as a reason to disengage (C4), and it recurs independently across multiple students, either as a disengagement trigger (S6) or as lived experience (S8, S11).

5.4.4. Outcome quality

Two professors report no formal mechanism for evaluating whether a project study delivered proportionate value (P1, P5); this connects directly to one company's independent complaint that project studies deliver limited usable insight (C2) and another professor's complaint that outcome quality is highly dependent on student engagement (P4).

5.4.5. Slow or unresponsive communication

Named as an application-ending concern by students (S8, S12, S13) and, from the opposite side of the same process, as staff's core administrative obstacle in obtaining professor responses (U2, U3).

6. Functional Requirements

6.1. Prioritization Scheme (MoSCoW, criteria defined here. Requirements grounded in cross-cutting pain points and structurally independent of TUM-specific administration are prioritized higher)

Requirements are prioritized using MoSCoW (Must, Should, Could, Won't), a judgement-based scheme, which is suited to the context of this project. Priority is assigned according to two criteria, applied in the following order: 1) Structural necessity. A requirement is Must-priority if the core application-to-supervision workflow cannot operate from start to finish without it. This holds true no matter how many interviews support it directly. 2) Strength of evidence. Among the requirements that are not structurally necessary, those founded on a cross-cutting finding or supported across multiple interviews are ranked as Should. Requirements based on a single citation, a team decision, or those that meet a narrower need are ranked as Could. Requirements that are clearly out of scope are marked as Won't in Section 4.6.

Transferability (Section 4.5) is noted for each requirement, but it does not influence this prioritization. The two attributes address different questions, and a requirement can do well in one without doing well in the other.

6.2. Access Features

6.2.1. FR-1, Role-Based Access (Must)

There are four different login/user types: Student, Company, Professor, and Staff. Each type has a unique dashboard. Although no interview explicitly requested this, every stakeholder-specific feature depends on it. Without separate roles, the platform cannot display different information to different users. This makes it a necessary requirement rather than an optional addition.

Source: Process Documentation (design decision, structural)

6.3. Company Features

6.3.1. FR-2, Company Project Submission Portal (Must)

The following fields are required: Project Title, Required Area of Expertise, Project Background and Objective, Project Deliverable, Available Company Resources, Required Student Skills, Group Size, Company Contact Person. This is adapted from

the current project proposal sheet.

Source: S7, S8, S9, P3, P4, P6

6.3.2. FR-3, Submission Review and Approval (Should)

New project submissions will be marked as 'pending'. Staff will approve them before professors can see them.

Source: Process Documentation

6.3.3. FR-4, Company Browse Student Topics (Must)

Companies can browse topics submitted by students. Acceptance requires the company to fill in mandatory fields (company name, company contact information), and the completed entry becomes visible to professors.

Source: C2 (directly suggested this feature), C5, S12

6.4. Professor Features

6.4.1. FR-5, Professor Profiles and Expertise Matching (Should)

Professor profiles will display their chair and area of expertise, sourced from institution logins. When logged in, professors will see project entries matched to their expertise.

Source: P3, P4, U1, U2

6.4.2. FR-6, Professor Supervision Take-on (Must)

To take on a project — either submitted by a company (FR-2) or a student topic accepted by a company (FR-4) — the professor provides chair contact information, application deadlines, and required documents. Once submitted, the project will be visible to students.

Source: Process Documentation

6.4.3. FR-7, Direct Project Submission (Should)

When a professor and a company agree directly on a project study, the professor will submit it with the required fields filled in for students to apply.

Source: Process Documentation

6.5. Student Features

6.5.1. FR-8, Student Project Submission Portal (Must)

The mandatory fields include: Project Title, Required Area of expertise, Project Background and Objective, Project Deliverable, Required Company Resources,

Group size, Student Team Contact Person (adapted from the current project proposal sheet).

Source: C2, C4, C5, S12

6.5.2. FR-9, Student Profiles (Should)

Student profiles will list their program or degree, sourced from institution logins. This will help match students with teams and allow professors and companies to filter students by their program.

Source: Team Decision

6.5.3. FR-10, Student Team Matching (Could)

A student can set their profile status to 'seeking teammates' and include a short message and contact information. Profiles can be filtered by this status.

Source: S10

6.5.4. FR-11, Student Browse Available Projects and Apply (Must)

Students can view available projects with complete information, including company details, expertise required, application deadlines, skills needed, and contact person. They can also apply for a given entry by submitting the required documents.

Source: S6, S7, S8, S9

6.5.5. FR-12, Application Status Visibility (Should)

Students will see a simple accepted/rejected status for each application.

Source: S1, S8, S12, S13

6.6. General Features

6.6.1. FR-13, Status-Change Notifications (Should)

Automated notifications, such as emails, will be sent when an application's status changes or when a listing closes.

Source: S8, U1, U2

6.6.2. FR-14, Lightweight Progress Check-in (Could)

A simple check-in mechanism will allow interaction between students and supervisors during the project.

Source: P4

6.6.3. FR-15, Dual Approval (Professor + Company) (Could)

A student's application will require approval from both the professor and the company before it is finalized. At TUM Campus Heilbronn, the selection of students is currently handled solely by the university, with no formal role for companies in reviewing individual applicants. This proposed change aims to give companies input on who joins their projects, which could lead to higher satisfaction with the outcomes. However, this change may slow down the process and add a coordination step between professors and companies, which are recognized costs of the benefit.

Source: Team Decision

6.6.4. FR-16, Manual Multi-Offer Decision (Should)

If a student is accepted to multiple projects, they will manually confirm one and withdraw from the others. This follows from FR-12 and FR-13: once the application statuses are real, a student accepting two projects simultaneously is a situation the platform must manage.

Source: Process Documentation

6.7. Administrative Staff Features

6.7.1. FR-17, Staff Master Dashboard (Should)

Staff will be able to view all project studies across every status: pending, approved, not yet supervised, ongoing, and filled.

Source: U1, U2

6.7.2. FR-18, Filter Incoming Submissions (Should)

Staff will have the ability to filter and search incoming company submissions during reviews. The volume of submissions will scale with FR-3. Filtering will ensure that manual reviews do not become bottlenecks.

Source: Process Documentation

7. Non-Functional Requirements

7.1. Usability

Four groups use the platform: students, companies, professors, and university staff. Not all of them work with software every day, professors and administrative staff least of all. Each role sees only the functions it needs (FR-1, Role-Based Access), so no one has to learn the parts of the system that belong to someone else. Every listing shows the same fields in the same order. Users compare projects, then find the same information in the same place each time. Inconsistent listings would bring back the fragmented presentation that the platform replaces.

7.1.1. NFR-3, Language Support

The platform is viewable in both English and German. German-language requirements were named as a disadvantage by international students (S5, S7, S10), particularly at institutions with a large international intake and degree programs taught in English.

Source: S5, S7, S10

7.2. Security and Data Privacy

7.2.1. NFR-1, Identity Verification

Use of institutional logins for verifying student and staff identities. A verified company email domain serves as a simple check for companies, before being made available to administrative staff for a more in-depth check.

Source: C4

7.2.2. NFR-2, Data Protection for Submitted Documents

CVs, transcripts, and motivation letters uploaded with an application (FR-11) count as personal data under the GDPR. Access is limited to the professor and the company that review the application. No other user can open these files or reach them through project listings. Student-submitted topics (FR-8) and the entries a company accepts (FR-4) stay visible to the other platform roles, but the documents attached to an application do not. How long these documents are stored, and when they are deleted once an application closes, lies outside the scope of this specification and the prototype (Section 4.6).

Source: Process Documentation (legal/regulatory constraint)

7.3. Maintainability / Extensibility

The requirements are structured so that Could-priority and out-of-scope features can be added later without changing the core Must-priority workflow. Student Team Matching (FR-10) and Lightweight Progress Check-in (FR-14) are extensions to the platform's functionality without requiring changes to it.

8. System Models

8.1. Use Case Diagram(s)

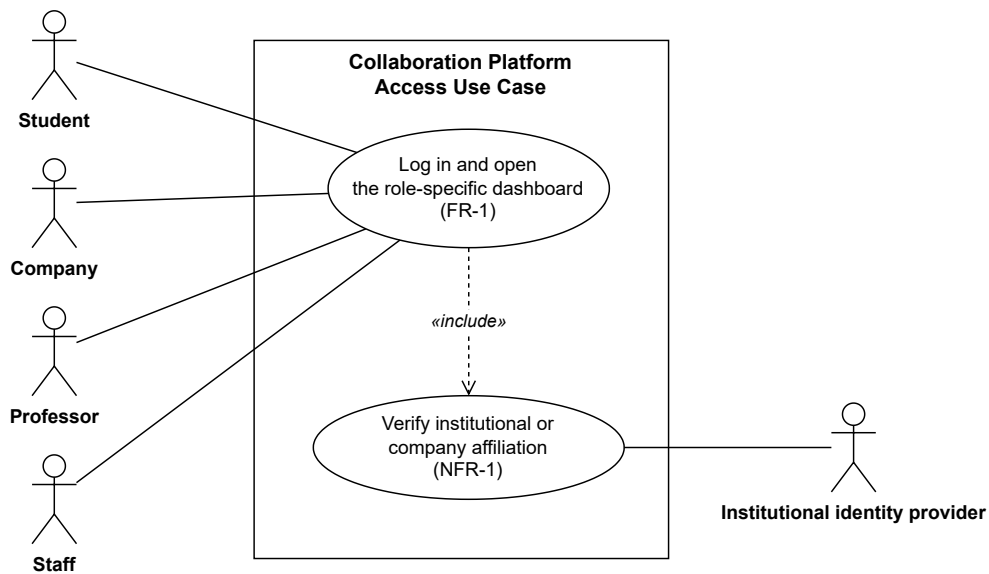


Figure 2: Platform access Use Case diagram

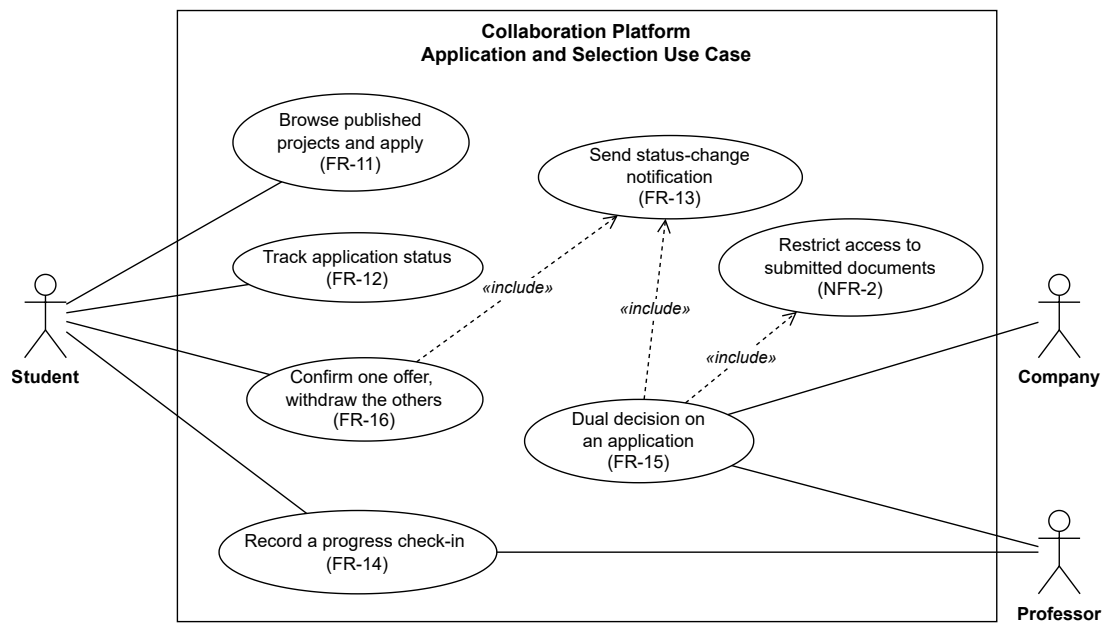


Figure 3: Application and Selection Use Case diagram

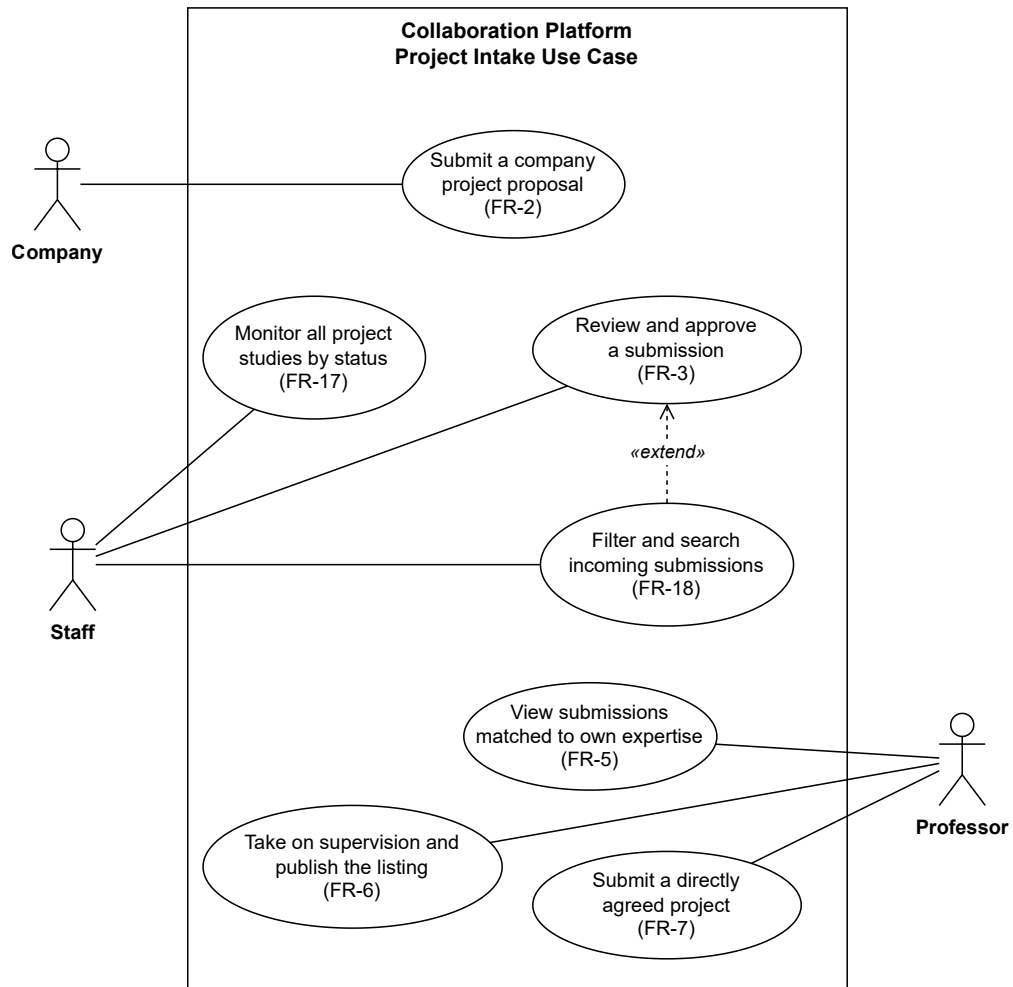


Figure 4: Project Intake Use Case diagram

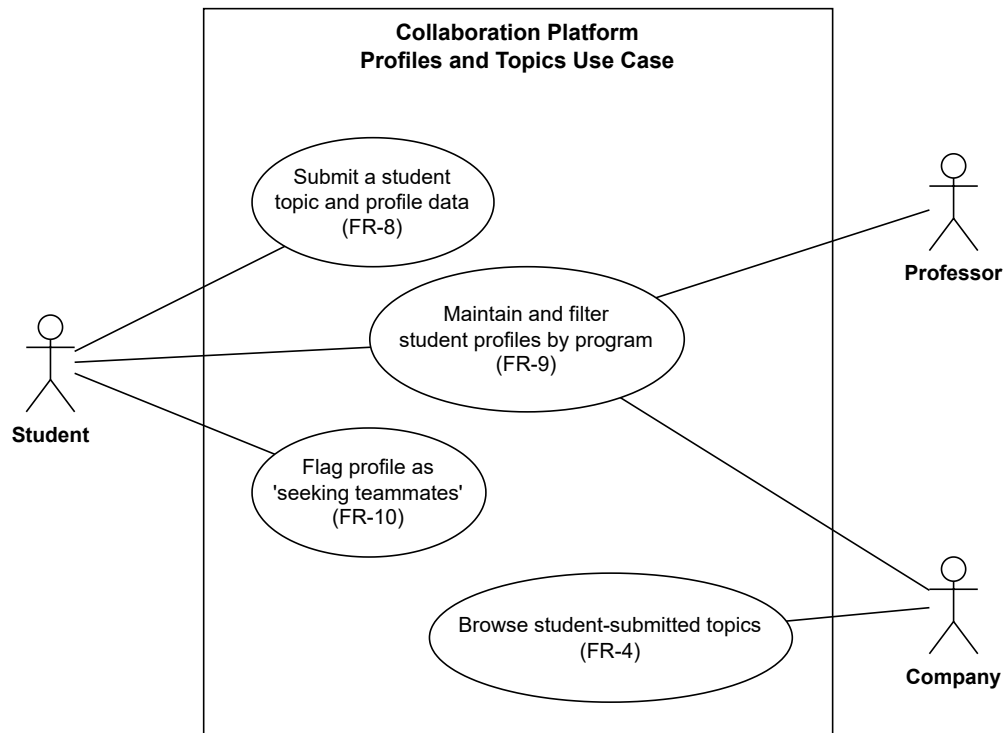


Figure 5: Profiles and Topics Use Case diagram

9. Traceability and Prototype Alignment

9.1. Traceability Table: Interview Insight → Pain Point → Requirement ID

The full traceability table, including prototype status, MoSCoW priority, and transferability tags for every requirement, is provided in Appendix C. The table below presents the abbreviated form for reference within the running text.

9.2. Prototype Coverage Table: Screen → Requirement ID

9.3. Pain Points Without Requirements and Justifications

The following pain points were identified but deliberately left without a corresponding requirement.

- **Language barrier** (S5, S7, S10): Partially addressed. Language Support (NFR-3) makes the platform itself usable in German and English, but the German-language requirements of the projects and chairs themselves lie outside what a platform can change.
- **Value and goal misalignment** (S6, S8, S11, U1, C2, C4): Dual Approval (FR-15) addresses company satisfaction with the selection, but not the underlying motivational mismatch, which cannot be addressed via a platform.
- **Outcome measurement** (P1, P5, C2): Defining an accurate quality measurement is highly complex.
- **Time and financial resource pressure** (P5): External constraint, not addressable via a platform.
- **University structural complexity and inconsistency across institutions** (C4): Acknowledged in the transferability discussion; not resolvable by a feature.
- **Output quality and software/tooling mismatches between student and company environments** (C2): Structural, and therefore largely not addressable via a platform. A further implementation of the light progress check (FR-14) could improve outcome quality by supporting more effective communication.
- **Screening signal degradation due to AI-generated CVs** (C3): Platform-side AI checks are technically conceivable, but their accuracy is low and the implementation effort is not proportionate to the value.

9.4. Requirements Without Prototype Coverage and Justifications

10. Risks, Discussion and Conclusion

This document has translated a set of semi-structured interviews across stakeholder groups into scoped requirements, which can be traced back to the corresponding pain point and traced forward to the prototype coverage, for a project studies platform specific to TUM Campus Heilbronn. This traceability, rather than the completeness of any one requirement, is the main assertion of this document.

Risks primarily rest on 26 interviews of uneven depth across groups, and especially since requirements were not checked back with the people they originated from. So the contents of this document are informed proposals, not validated findings. Two points follow from that directly. Company and professor findings come from smaller, less saturated samples than the student group, so they're less likely to hold up even within TUM Campus Heilbronn specifically. And Dual Approval (FR-15) is a proposed change, not something already happening. The reasoning behind it (companies having more of a say in the process improves satisfaction with outcomes) holds up logically, but nothing actually tests whether it would be effective or introduce further complications. One more risk is not related to the interviews at all: the platform handles personal application data (Section 7.2), and no one with legal expertise has reviewed what that would require.

The transferability tags in the traceability table give pointers on how this work can be extended beyond TUM Campus Heilbronn, but that's discussed in principle in Section 4.5 and nothing more, it hasn't been tried. Closer to what's actually in hand are the broader industry-academia findings collected: experiences, motivations, barriers, what keeps people from collaborating in the first place.

References

- Achimugu, P., Selamat, A., Ibrahim, R., & Mahrin, M. (2014). A systematic literature review of software requirements prioritization research. *Information and Software Technology*, 56, . <https://doi.org/10.1016/j.infsof.2014.02.001>
- Ankrah, S., & AL-Tabbaa, O. (2015). Universities-industry collaboration: A systematic review. *Scandinavian Journal of Management*, 31(3), 387–408. <https://doi.org/https://doi.org/10.1016/j.scaman.2015.02.003>
- Bruneel, J., D'Este, P., & Salter, A. (2010). Investigating the factors that diminish the barriers to university-industry collaboration. *Research Policy*, 39(7), 858–868. <https://doi.org/https://doi.org/10.1016/j.respol.2010.03.006>
- Ferrari, A., Spoletini, P., & Debnath, S. (2022,). *How Do Requirements Evolve During Elicitation? An Empirical Study Combining Interviews and App Store Analysis*. <https://arxiv.org/abs/2208.00825>
- Gotel, O., & Finkelstein, A. (1994). An analysis of the requirements traceability problem. *Proceedings of IEEE International Conference on Requirements Engineering*, 94–101. <https://api.semanticscholar.org/CorpusID:5870868>
- Guest, G., Bunce, A., & Johnson, L. (2006). How Many Interviews Are Enough?: An Experiment with Data Saturation and Variability. *Field Methods*, 18, 59–82. <https://doi.org/10.1177/1525822X05279903>
- Hagaman, A. K., & Wutich, A. (2017). How Many Interviews Are Enough to Identify Metathemes in Multisited and Cross-cultural Research? Another Perspective on Guest, Bunce, and Johnson's (2006) Landmark Study. *Field Methods*, 29(1), 23–41. <https://doi.org/10.1177/1525822X16640447>
- Hennink, M. M., Kaiser, B. N., & Marconi, V. C. (2017). Code Saturation Versus Meaning Saturation: How Many Interviews Are Enough?. *Qualitative Health Research*, 27(4), 591–608. <https://doi.org/10.1177/1049732316665344>
- Howells, J. (2006). Intermediation and the role of intermediaries in innovation. *Research Policy*, 35(5), 715–728. <https://doi.org/https://doi.org/10.1016/j.respol.2006.03.005>
- MacQueen, K. M., McLellan, E., Kay, K., & Milstein, B. (1998). Codebook Development for Team-Based Qualitative Analysis. *CAM Journal*, 10(2), 31–36. <https://doi.org/10.1177/1525822X980100020301>

- Perkmann, M., Tartari, V., McKelvey, M., Autio, E., Broström, A., D'Este, P., Fini, R., Geuna, A., Grimaldi, R., Hughes, A., Krabel, S., Kitson, M., Llerena, P., Lissoni, F., Salter, A., & Sobrero, M. (2013). Academic engagement and commercialisation: A review of the literature on university-industry relations. *Research Policy*, 42(2), 423–442. <https://doi.org/https://doi.org/10.1016/j.respol.2012.09.007>
- Society, I. C. (1998). *IEEE 830-1998;IEEE Recommended Practice for Software Requirements Specifications*. <https://standards.ieee.org/ieee/830/1222/>
- Society, I. C. (2018). *ISO/IEC/IEEE 29148:2018; ISO/IEC/IEEE International Standard - Systems and software engineering – Life cycle processes – Requirements engineering*. <https://standards.ieee.org/ieee/29148/6937/>

Appendices

Interview ID	Interviewee	Stakeholder Group
S1	TUM Campus Garching Student 1	Student
S2	TUM Campus Heilbronn Student 1	Student
S3	TUM Campus Garching Student 2	Student
S4	TUM Campus Heilbronn Student 2	Student
S5	TUM Campus Heilbronn Student 3	Student
S6	RWTH Aachen University Student	Student
S7	TUM Campus Heilbronn Student 4	Student
S8	TUM Campus Heilbronn Student 5	Student
S9	TUM Campus Heilbronn Student 6	Student
S10	TUM Campus Heilbronn Student 7	Student
S11	Reutlingen University Student	Student
S12	TUM Campus Heilbronn Student 8	Student
S13	TUM Campus Heilbronn Student 9	Student
C1	Founder of IT Solution Startup	Company
C2	IDS HR Business Partner	Company
C3	Atlas Copco Manager Apprenticeship	Company
C4	SAP Campus Ambassadors Lead	Company
C5	Schneider HR Department	Company
P1	Koblenz University Professor	Professor (external)
P2	TUM Campus Munich Professor	Professor (external, other campus)
P3	TUM Campus Heilbronn Professor	Professor (TUM Heilbronn)
P4	TUM Campus Heilbronn Professor	Professor (TUM Heilbronn)
P5	RWTH Aachen University Professor	Professor (external)
P6	Passau University PhD Candidate	Research Candidate (external)
U1	TUM Campus Heilbronn Administrative Staff 1	University staff (TUM Heilbronn)
U2	TUM Campus Heilbronn Administrative Staff 2	University staff (TUM Heilbronn)

Table 4: Interview Sample Composition and Participant IDs ($n = 26$)